

OBJECT:

- To determine your own bleeding times by Duke's method.

OBJECTIVES:

The students should be able to:

Learn the normal bleeding time and causes of prolonged bleeding time

- The student should be able to:
 - Find out the clotting time.
 - Know the causes which may lead to prolonged and defective clotting time.
 - Have knowledge about anticoagulants used in the laboratory.
 - Understand the:
 - 1) Hemophilia
 - 2) Thrombosis
 - 3) Thromboembolic condition and their hazards
 - 4) Differentiate between bleeding time and clotting time.

REQUIREMENTS:

- Filter paper,
- Stop watch and
- All the material for pricking the finger.

THEORETICAL EXPLANATION:

- When ever there is a trauma, wound, cut or injury bleeding takes place
- To prevent the excessive loss of blood, nature has provided the mechanism of “**hemostasis**” that occurs in different steps
- The first step is local **myogenic effect** which is prolonged by nervous control
- The second step is the formation of **platelet plug**

THEORETICAL EXPLANATION:

- As the vessel is traumatized, the nearby passing platelets come and adhere to the **rough surface (collagen fibers)** and secrete **ADPs** and **thromboxane A_2** which make these platelets sticky for other passing platelets i.e., a platelet plug is formed which plugs the gap in the vessel
- The sharply cut vessel bleeds more than crushed vessel
- The more a vessel is **traumatized**, more is the **degree of spasm**
- Platelet plug is unstable and temporary .proper hemostasis occurs due to addition of clotting factors leading to formation of fibrin.

PRINCIPLE OF EXPERIMENT:

- Principle of this experiment is to note the time when bleeding stops due to formation of platelet plug
- The bleeding time is taken for a small sharp incision to stop bleeding

PROCEDURE:

- After all usual precautionary measures, prick the finger with pre-sterilized disposable blood lancet about 3mm deep and start the stop watch
- Wipe off the blood from the site
- Apply a gentle pressure and touch the site of prick with filter paper so that the spot of blood appears on the filter paper.
- Repeat the procedure after every 30 sec by touching the site of prick with filter paper and wiping it off every time till there is no spot of blood on filter paper.

PROCEDURE

- Stop the stop watch
- Circle the spots of blood with pencil and note the time of each spot
- Note the bleeding time and paste the filter paper on your journal
- Bleeding time is also calculated by dividing total no. of drops on the filter paper by two
- **Bleeding time = No. of drops on the filter paper / 2**

CLINICAL SIGNIFICANCE:

- There are various conditions in which the bleeding time is prolonged
- Thrombocytopenic purpura is the most t important
- Non Thrombocytopenic purpura are also there due to:
- Vitamin C deficiency (**scurvy**)
- Infection leading to consumption of clotting factors & deficiency of platelet
- Allergic reactions
- Autoimmune reactions
- Idiopathic thrombocytopenia
- Aspirin inhibit platelet aggregation inhibiting the formation of prostaglandin and thromboxane. It rarely causes bleeding tendency because it does not inhibit aggregation to sufficient degree
- Thrombocytopenic purpura result when total platelet count falls below 50,000/cu.mm of blood

CLINICAL SIGNIFICANCE:

- Normal bleeding time by Duke's method is 2 to 6 min
- However it depends, on various factors e.g.,
- Depth of wound, cut or injury.
- Degree of hyperemia in the finger
- Total platelet count in blood.
- Elasticity and the condition of skin.
- Several substances are secreted at the site of pricks e.g.,
Serotonin,
- **thromboxaneA₂,**
- **5-hydroxytryptophanamine (5HT), histamine**

THROMBOCYTOPENIC PURPURA:

- Bleeding disorder which is caused by lack of platelets when their count is decrease below 50.000/cu.mm. In this, the bleeding time is prolonged but the clotting time remains normal
- The blood oozes out of the blood vessels and clot forming blue patches called purpura
- Subcutaneously or with in the tissues
- Normally the blood vessels are damaged several times a day and repaired at the same
- Time by platelet plugs

PRECAUTIONS:

- No need to discard first drop of blood
- The prick should be 3mm deep
- Do not squeeze the finger, apply only a gentle pressure
- Start the stop watch as you prick the finger

OBSERVATIONS:

- **RESULT:**

- My bleeding time is _____ sec.

Relevant Questions

Q-1-What is Bleeding time?

Bleeding time is a medical test that measures how long it takes for a small, standardized cut on the skin to stop bleeding.

Q-2-What is normal value?

The normal bleeding time is usually between 2 and 7 minutes.

Q-3-What is the role of platelets to stop the bleeding?

Platelets, also known as thrombocytes, are crucial for stopping bleeding by forming a blood clot at the site of an injury.

Q-4-What are the clinical conditions in which bleeding time is prolonged?

Diseases that may cause prolonged bleeding time include thrombocytopenia,

disseminated intravascular coagulation (DIC),

Bernard-Soulier disease, and Glanzmann's thrombasthenia.

Aspirin and other cyclooxygenase inhibitors can significantly prolong bleeding time

Q-5-Is there any other method other than Duke's for determination of bleeding time?

The Ivy Lee method is a simple yet effective time management technique that was developed in the early 20th century by productivity consultant, Ivy Lee